

# Model Question Paper-I with effect from 2025

USN

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1BMATM201

## Second Semester B.E./ B.Tech. Degree Examination Multivariable Calculus and Numerical Methods

TIME:03 Hours

Max.Marks:100

- Note: 1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**  
 2. VTU Formula Hand Book is Permitted  
 3. M: Marks, L: Bloom's level, C: Course outcomes

| Module-1 |   |                                                                                                                                      | M | L  | C |
|----------|---|--------------------------------------------------------------------------------------------------------------------------------------|---|----|---|
| Q 1.     | a | Evaluate $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dz dy dx$ .                                                          | 6 | L2 | 1 |
|          | b | Evaluate $\int_0^a \int_y^a \frac{x}{x^2+y^2} dz dx dy$ by changing the order of integration.                                        | 7 | L2 | 1 |
|          | c | Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ and show that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$        | 7 | L2 | 1 |
| OR       |   |                                                                                                                                      |   |    |   |
| Q 2.     | a | Evaluate $\int_{-1}^{+1} \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$                                                             | 6 | L2 | 1 |
|          | b | Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates.                                        | 7 | L2 | 1 |
|          | c | Show that $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta = \pi$ | 7 | L2 | 1 |
| Module-2 |   |                                                                                                                                      |   |    |   |
| Q 3.     | a | Solve $\frac{d^2y}{dx^2} - 4y = \cosh(2x - 1) + 3^x$ .                                                                               | 6 | L2 | 2 |
|          | b | Solve $\frac{d^2y}{dx^2} + y = \sec x \tan x$ by the method of variation of parameters                                               | 7 | L2 | 2 |
|          | c | Solve $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 4y = (1+x)^2$                                                                      | 7 | L2 | 2 |
| OR       |   |                                                                                                                                      |   |    |   |
| Q 4.     | a | Solve $\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 2y = 4 \cos^2 x$ .                                                                      | 6 | L2 | 2 |
|          | b | Using the method of variation of parameters<br>Solve $\frac{d^2y}{dx^2} + y = \frac{1}{1+\sin x}$                                    | 7 | L2 | 2 |
|          | c | Solve $(1+x)^2 y'' + (1+x)y' + y = 2 \sin \log(1+x)$                                                                                 | 7 | L2 | 2 |
| Module-3 |   |                                                                                                                                      |   |    |   |
| Q 5.     | a | Find $\text{div} \vec{F}$ & $\text{Curl} \vec{F}$ where $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$ .                                 | 6 | L2 | 3 |

|          |                                                                                                                                                       |                                                                                                                                                                                                                                        |    |    |    |   |    |   |    |    |    |    |   |    |   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|---|----|---|----|----|----|----|---|----|---|
|          | <b>b</b>                                                                                                                                              | Find the value of the constant 'a' such that the vectorfield , $\vec{F} = (axy - z^3)\mathbf{i} + (a - 2)x^2\mathbf{j} + (1 - a)xz^2\mathbf{k}$ is irrotational & hence find a scalar function $\Phi$ such that $\vec{F} = \nabla\Phi$ | 7  | L2 | 3  |   |    |   |    |    |    |    |   |    |   |
|          | <b>c</b>                                                                                                                                              | Verify Green's theorem for $\int (xy + y^2)dx + x^2dy$ .where C is Closed curve of the region bounded by $y = x$ & $y = x^2$ .                                                                                                         | 7  | L2 | 3  |   |    |   |    |    |    |    |   |    |   |
| OR       |                                                                                                                                                       |                                                                                                                                                                                                                                        |    |    |    |   |    |   |    |    |    |    |   |    |   |
| Q 6.     | <b>a</b>                                                                                                                                              | If $\vec{F} = \nabla(xy^3z^2)$ find $\text{div}\vec{F}$ & $\text{curl}\vec{F}$ at the point (1, -1 , 1) .                                                                                                                              | 6  | L2 | 3  |   |    |   |    |    |    |    |   |    |   |
|          | <b>b</b>                                                                                                                                              | If $\vec{F} = (x+y+az)\mathbf{i} + (bx + 2y - z)\mathbf{j} + (x + cy + 2z)\mathbf{k}$ Find a, b, c such that $\text{curl}\vec{F} = 0$ & then find $\Phi$ such that $\vec{F} = \nabla\Phi$ .                                            | 7  | L3 | 3  |   |    |   |    |    |    |    |   |    |   |
|          | <b>c</b>                                                                                                                                              | Verify stoke's theorem for vector $\vec{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$ taken round the rectangle by $x = 0, x = a, y = 0, y = b$ .                                                                                        | 7  | L2 | 3  |   |    |   |    |    |    |    |   |    |   |
| Module-4 |                                                                                                                                                       |                                                                                                                                                                                                                                        |    |    |    |   |    |   |    |    |    |    |   |    |   |
| Q 7.     | <b>a</b>                                                                                                                                              | Find a real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places, by using Regula-falsi method                                                                                                                      | 6  | L2 | 4  |   |    |   |    |    |    |    |   |    |   |
|          | <b>b</b>                                                                                                                                              | Given $f(40) = 184, f(50) = 204, f(60) = 226, f(70) = 250, f(80) = 276, f(90) = 304$ , find $f(38)$ & $f(85)$ using suitable interpolation formulae.                                                                                   | 7  | L2 | 4  |   |    |   |    |    |    |    |   |    |   |
|          | <b>c</b>                                                                                                                                              | Evaluate $\int_0^1 \frac{x dx}{1+x^2}$ by Trapezoidal rule taking seven ordinates & hence find $\log_e 2$ .                                                                                                                            | 7  | L2 | 4  |   |    |   |    |    |    |    |   |    |   |
| OR       |                                                                                                                                                       |                                                                                                                                                                                                                                        |    |    |    |   |    |   |    |    |    |    |   |    |   |
| Q 8.     | <b>a</b>                                                                                                                                              | Find the real root of the Equation $xe^x - 2 = 0$ correct to three decimal places by applying Newton Raphson method.                                                                                                                   | 6  | L2 | 4  |   |    |   |    |    |    |    |   |    |   |
|          | <b>b</b>                                                                                                                                              | Use Lagrange's interpolation formula to find y at x=10 give<br><table><tr><td>x</td><td>5</td><td>6</td><td>9</td><td>11</td></tr><tr><td>y</td><td>12</td><td>13</td><td>14</td><td>16</td></tr></table>                              | x  | 5  | 6  | 9 | 11 | y | 12 | 13 | 14 | 16 | 7 | L2 | 4 |
|          | x                                                                                                                                                     | 5                                                                                                                                                                                                                                      | 6  | 9  | 11 |   |    |   |    |    |    |    |   |    |   |
| y        | 12                                                                                                                                                    | 13                                                                                                                                                                                                                                     | 14 | 16 |    |   |    |   |    |    |    |    |   |    |   |
| <b>c</b> | Use simpson's $\frac{3^{th}}{8}$ rule to Obtain the approximate value of $\int_0^{0.3} (1 - 8x^3)^{\frac{1}{2}} dx$ by considering 3 equal intervals. | 7                                                                                                                                                                                                                                      | L2 | 4  |    |   |    |   |    |    |    |    |   |    |   |
| Module-5 |                                                                                                                                                       |                                                                                                                                                                                                                                        |    |    |    |   |    |   |    |    |    |    |   |    |   |
| Q 9.     | <b>a</b>                                                                                                                                              | Use Taylor's series method to find y at $x = 0.1, 0.2, 0.3$ considering terms upto the third degree given that $\frac{dy}{dx} - 2y = x^2 + y^2$ & $y(0) = 1$ .                                                                         | 6  | L2 | 5  |   |    |   |    |    |    |    |   |    |   |
|          | <b>b</b>                                                                                                                                              | Given $\frac{dy}{dx} = 1 + \frac{y}{x}, y = 2$ , at $x = 1$ , find the approximate value y at $x = 1.4$ by taking step size $h = 0.2$ applying modified Euler's method.                                                                | 7  | L2 | 5  |   |    |   |    |    |    |    |   |    |   |
|          | <b>c</b>                                                                                                                                              | Given that $\frac{dy}{dx} = x - y^2$ and the data $y(0) = 0, y(0.2) = 0.02, y(0.4) = 0.0795, y(0.6) = 0.1762$ . Compute $y(0.8)$ by applying Milne's method.                                                                           | 7  | L2 | 5  |   |    |   |    |    |    |    |   |    |   |
| OR       |                                                                                                                                                       |                                                                                                                                                                                                                                        |    |    |    |   |    |   |    |    |    |    |   |    |   |
| Q 10.    | <b>a</b>                                                                                                                                              | Employ Talyor's series method to find y at $x = 0.1$ given $\frac{dy}{dx} - 2y = 3e^x$ Whose solution passes through the origin.                                                                                                       | 6  | L2 | 5  |   |    |   |    |    |    |    |   |    |   |
|          | <b>b</b>                                                                                                                                              | Using Runge-Kutta method of fourth order, find $y(0.2)$ for the equation $\frac{dy}{dx} = \frac{y - x}{y + x}, y(0) = 1$ taking $h = 0.2$                                                                                              | 7  | L2 | 5  |   |    |   |    |    |    |    |   |    |   |
|          | <b>c</b>                                                                                                                                              | If $\frac{dy}{dx} = 2e^x - y, y(0) = 2, y(0.1) = 2.010, y(0.2) = 2.040$ & $y(0.3) = 2.090$ , find $y(0.4)$ Using Milne's method.                                                                                                       | 7  | L3 | 5  |   |    |   |    |    |    |    |   |    |   |

# CBCS SCHEME

USN

1V525ME004

1BMATM201

## Second Semester B.E./B.Tech. Degree Examination, June/July 2026 Multivariable Calculus and Numerical Methods : ME Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks, L: Bloom's level, C: Course outcomes.  
3. VTU Formula Hand Book is permitted.

| Module - 1 |    |                                                                                                                             | M | L  | C   |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Evaluate the integral $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dz dy dx$                                      | 7 | L3 | CO1 |
|            | b. | Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy dy dx$ by changing the order of the integration.                                    | 7 | L3 | CO1 |
|            | c. | Prove the Relationship between Beta and Gamma functions :<br>$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$         | 6 | L3 | CO1 |
| OR         |    |                                                                                                                             |   |    |     |
| Q.2        | a. | Evaluate $\iint xy(x+y) dy dx$ taken over the area between $y = x^2$ to $y = x$                                             | 7 | L3 | CO1 |
|            | b. | Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4, z = 0$                                   | 7 | L3 | CO1 |
|            | c. | Show that $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin\theta}} \int_0^{\frac{\pi}{2}} \sqrt{\sin\theta} d\theta = \pi$ | 6 | L3 | CO1 |
| Module - 2 |    |                                                                                                                             |   |    |     |
| Q.3        | a. | Solve the D.E. $\frac{d^3 y}{dx^3} + 6\frac{d^2 y}{dx^2} + 11\frac{dy}{dx} + 6y = 0$                                        | 6 | L3 | CO1 |
|            | b. | Solve $\frac{d^2 y}{dx^2} - 4y = \cosh(2x - 1) + 3^x$                                                                       | 7 | L3 | CO1 |
|            | c. | Solve $x^2 y'' - 3xy' + 4y = (1+x)^2$                                                                                       | 7 | L3 | CO1 |
| OR         |    |                                                                                                                             |   |    |     |
| Q.4        | a. | Solve $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$                                                                    | 6 | L3 | CO1 |
|            | b. | Solve: $y'' + 3y' + 2y = 12x^2$                                                                                             | 7 | L3 | CO1 |
|            | c. | Solve $(2x+1)^2 y'' - 6(2x+1)y' + 16y = 8(2x+1)^2$                                                                          | 7 | L3 | CO1 |
| 1 of 3     |    |                                                                                                                             |   |    |     |

Module - 3

|     |    |                                                                                                                                           |   |    |     |
|-----|----|-------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.5 | a. | If $\vec{F} = xyz\hat{i} + 3x^2y\hat{j} + (xz^2 - y^2z)\hat{k}$ , find the $\text{div}\vec{F}$ and $\text{curl}\vec{F}$ at $(2, -1, 1)$ . | 6 | L3 | CO1 |
|     | b. | Find the directional derivative of the function $\phi = xy^2 + yz^3$ at $(2, -1, 1)$ along $\hat{i} + 2\hat{j} + 3\hat{k}$                | 7 | L3 | CO1 |
|     | c. | Find the angle between the surfaces or normal surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$ .                    | 7 | L3 | CO1 |

OR

|     |    |                                                                                                                                                                                                           |   |    |     |
|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.6 | a. | P.T $\vec{F} = (y + z)\hat{i} + (z + x)\hat{j} + (x + y)\hat{k}$ is irrotational. Also find a scalar point function $\phi$ such that $\vec{F} = \nabla\phi$ .                                             | 7 | L3 | CO1 |
|     | b. | If $\vec{F} = (3x^2 + 6y)\hat{i} - (14yz)\hat{j} + (20xz^2)\hat{k}$ , evaluate $\int_C \vec{F} \cdot d\vec{r}$ from the point $(0,0,0)$ to $(1,1,1)$ along the curve given by $x = t, y = t^2, z = t^3$ . | 7 | L3 | CO1 |
|     | c. | Evaluate $\int_C (xy - x^2)dx + x^2ydy$ where C is the closed curve bounded by $y = 0, x = 1$ and $y = x$ .                                                                                               | 6 | L3 | CO1 |

Module - 4

|            |                                                                                                                                                                                                          |                                                                                                                                                                     |       |       |       |   |    |     |      |      |      |      |      |      |            |       |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|---|----|-----|------|------|------|------|------|------|------------|-------|
|            |                                                                                                                                                                                                          | Module - 4                                                                                                                                                          |       |       |       | 6 | L3 | CO2 |      |      |      |      |      |      |            |       |
| Q.7        | a.                                                                                                                                                                                                       | Find the real root of the equation $x \log_{10} x = 1.2$ by Regula falsi method, Carry out 3 iterations.                                                            |       |       |       | 7 | L3 | CO2 |      |      |      |      |      |      |            |       |
|            | b.                                                                                                                                                                                                       | The population of a town is given by the following data                                                                                                             |       |       |       |   |    |     |      |      |      |      |      |      |            |       |
|            | <table><tr><td>Year</td><td>1971</td><td>1981</td><td>1991</td><td>2001</td><td>2011</td></tr><tr><td>Population</td><td>19.96</td><td>39.65</td><td>58.81</td><td>77.18</td><td>94.58</td></tr></table> |                                                                                                                                                                     |       |       |       |   |    |     | Year | 1971 | 1981 | 1991 | 2001 | 2011 | Population | 19.96 |
| Year       | 1971                                                                                                                                                                                                     | 1981                                                                                                                                                                | 1991  | 2001  | 2011  |   |    |     |      |      |      |      |      |      |            |       |
| Population | 19.96                                                                                                                                                                                                    | 39.65                                                                                                                                                               | 58.81 | 77.18 | 94.58 |   |    |     |      |      |      |      |      |      |            |       |
|            |                                                                                                                                                                                                          | Using appropriate interpolation formula calculate the increase in the population from the year 1975 to 2005.                                                        |       |       |       |   |    |     |      |      |      |      |      |      |            |       |
|            | c.                                                                                                                                                                                                       | Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Simpson's $\frac{1^{rd}}{3}$ rule by taking 5 ordinates (4 equal parts) and hence deduce an approximate value of $\pi$ . |       |       |       | 7 | L3 | CO2 |      |      |      |      |      |      |            |       |

OR

| OR   |                                                                                                         |                                                                                                                                                                                                                                                   |    |     |     |   |   |   |      |   |   |    |     |     |   |    |     |
|------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|---|---|---|------|---|---|----|-----|-----|---|----|-----|
| Q.8  | a.                                                                                                      | Use Newton Raphson method to find the real root of the equation $xe^x = 2$ , correct to 3 decimal places.                                                                                                                                         | 7  | L3  | CO2 |   |   |   |      |   |   |    |     |     |   |    |     |
|      | b.                                                                                                      | Find $f(0.05)$ using Newton divided difference formula given that<br><table border="1"><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>5</td><td>6</td></tr><tr><td>f(x)</td><td>0</td><td>6</td><td>21</td><td>105</td><td>186</td></tr></table> | x  | 0   | 2   | 3 | 5 | 6 | f(x) | 0 | 6 | 21 | 105 | 186 | 7 | L3 | CO2 |
|      | x                                                                                                       | 0                                                                                                                                                                                                                                                 | 2  | 3   | 5   | 6 |   |   |      |   |   |    |     |     |   |    |     |
| f(x) | 0                                                                                                       | 6                                                                                                                                                                                                                                                 | 21 | 105 | 186 |   |   |   |      |   |   |    |     |     |   |    |     |
| c.   | Evaluate $\int_4^{5.2} \log_e x \, dx$ using Simpson's $\frac{3^{th}}{8}$ rule by taking 6 equal parts. | 6                                                                                                                                                                                                                                                 | L3 | CO2 |     |   |   |   |      |   |   |    |     |     |   |    |     |

## Module - 5

|      |    |                                                                                                                                                                                                                                                        |   |    |     |
|------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.9  | a. | Use Milne's predictor-corrector method. Given the differential equation, given $\frac{dy}{dx} = x^2(1+y)$ , find $y$ at $x=1.4$ . Carry out two corrections for the solution, given that $y(1)=1$ , $y(1.1)=1.233$ , $y(1.2)=1.548$ , $y(1.3)=1.979$ . | 7 | L3 | CO2 |
|      | b. | Using the modified Euler's method, solve the IVP $\frac{dy}{dx} = x + y^2$ , $y(0) = 1$ at points $x=0.1$ in steps of length $h=0.1$ . Carry out two modifications.                                                                                    | 6 | L3 | CO2 |
|      | c. | Using Runge Kutta method of fourth order solve $\frac{dy}{dx} = \frac{(y^2-x^2)}{(y^2+x^2)}$ for $x = 0.2$ given that $y = 1$ at $x = 0$ initially, by applying Runge - Kutta method of order 4.                                                       | 7 | L3 | CO2 |
| OR   |    |                                                                                                                                                                                                                                                        |   |    |     |
| Q.10 | a. | Using the Taylor's series method, find an approximate solution correct to four decimals at $x=0.1$ for the IVP $\frac{dy}{dx} = 2y + 3e^x$ , $y(0) = 0$                                                                                                | 6 | L3 | CO2 |
|      | b. | Using RK method of fourth order, to find $y(0.1)$ , given that $\frac{dy}{dx} = x^2 - y$ and $y(0) = 1$ take $h = 0.1$                                                                                                                                 | 7 | L3 | CO2 |
|      | c. | Use Adams -Bash forth method to find the value of $y$ at $x=0.8$ , given $\frac{dy}{dx} = x - y^2$ , where $y(0)=0$ , $y(0.2)=0.02$ , $y(0.4)=0.0795$ , $y(0.6)=0.1762$ . Apply corrector formula twice.                                               | 7 | L3 | CO2 |

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**Second Semester B.E./B.Tech. Degree Examination, June/July 2026**  
**Multivariable Calculus and Numerical Methods : ME**  
**Stream**

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks, L: Bloom's level, C: Course outcomes.

3. VTU Formula Hand Book is permitted.

| Module - 1 |    |                                                                                                                                                        | M | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dy dx$                                                                                                | 6 | L2 | CO1 |
|            | b. | Change the order of integration and hence Evaluate $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} xy dy dx$                                                   | 7 | L2 | CO1 |
|            | c. | Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$                                                                                      | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                                        |   |    |     |
| Q.2        | a. | Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$                                                                                  | 6 | L2 | CO1 |
|            | b. | Evaluate $\int_{-a}^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2 + y^2} dy dx$ by changing into polar coordinates.                                              | 7 | L2 | CO1 |
|            | c. | Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.                                                            | 7 | L3 | CO1 |
| Module - 2 |    |                                                                                                                                                        |   |    |     |
| Q.3        | a. | Solve $(D^5 - D^4 - D + 1)y = 0$                                                                                                                       | 6 | L2 | CO1 |
|            | b. | Solve $(D^3 + D^2 + 4D + 4)y = x^2 - 4x - 6$                                                                                                           | 7 | L3 | CO1 |
|            | c. | Solve $x^3 \frac{d^3 y}{dx^3} + 3x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 8y = 65 \cos(\log x)$                                                      | 7 | L3 | CO1 |
| OR         |    |                                                                                                                                                        |   |    |     |
| Q.4        | a. | Solve $(6D^2 + 17D + 12)y = e^{-x}$                                                                                                                    | 6 | L2 | CO1 |
|            | b. | Solve by the method of variation of parameters $\frac{d^2 y}{dx^2} + a^2 y = \sec ax$                                                                  | 7 | L3 | CO1 |
|            | c. | Solve $(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = \sin 2[\log(1+x)]$                                                                       | 7 | L3 | CO1 |
| Module - 3 |    |                                                                                                                                                        |   |    |     |
| Q.5        | a. | Find the directional derivative of $\phi = x^2 yz + 4xz^2$ at $(1, -2, -1)$ in the direction of $2i - j - 2k$                                          | 6 | L2 | CO1 |
|            | b. | If $\vec{F} = (3x^2 y - z)i + (xz^3 + y^4)j - 2x^3 z^2 k$ , find $\text{grad}(\text{div } \vec{F})$ at $(2, -1, 0)$                                    | 7 | L2 | CO1 |
|            | c. | Apply Green's theorem to evaluate $\int (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region enclosed by $y = \sqrt{x}$ and $y = x^2$ | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                                        |   |    |     |
| Q.6        | a. | Find the angle between the normals to the surface $xy = z^2$ at the points $(4, 1, 2)$ and $(3, 3, -3)$                                                | 6 | L2 | CO1 |

|    |                                                                                                                                                                                             |   |    |     |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| b. | Show that $\vec{F} = (y+z)\vec{i} + (z+x)\vec{j} + (x+y)\vec{k}$ is irrotational. Also find a scalar function $\phi$ such that $\vec{F} = \nabla\phi$                                       | 7 | L2 | CO1 |
| c. | If $\vec{F} = (3x^2 + 6y)\vec{i} - 14yz\vec{j} + 20xz^2\vec{k}$ , evaluate $\int \vec{F} \cdot d\vec{r}$ from (0,0,0) to (1,1,1) along the curve given by $x = t$ , $y = t^2$ , $z = t^3$ . | 7 | L2 | CO1 |

## Module - 4

|     |      |                                                                                                                                                                                                                                                                                                                                                                                              |      |      |      |    |    |     |   |      |      |      |      |      |   |    |     |
|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|----|----|-----|---|------|------|------|------|------|---|----|-----|
| Q.7 | a.   | Find the real root of the equation $\cos x = 3x - 1$ correct to three decimals by using regula falsi method.                                                                                                                                                                                                                                                                                 | 6    | L2   | CO2  |    |    |     |   |      |      |      |      |      |   |    |     |
|     | b.   | The area of a circle (A) corresponding to diameter (D) is given below. Find the area corresponding to diameter 105 using an appropriate interpolation formula,<br><table border="1" data-bbox="327 560 981 649"> <tr> <td>D</td><td>80</td><td>85</td><td>90</td><td>95</td><td>100</td></tr> <tr> <td>A</td><td>5026</td><td>5674</td><td>6362</td><td>7088</td><td>7854</td></tr> </table> | D    | 80   | 85   | 90 | 95 | 100 | A | 5026 | 5674 | 6362 | 7088 | 7854 | 7 | L3 | CO2 |
| D   | 80   | 85                                                                                                                                                                                                                                                                                                                                                                                           | 90   | 95   | 100  |    |    |     |   |      |      |      |      |      |   |    |     |
| A   | 5026 | 5674                                                                                                                                                                                                                                                                                                                                                                                         | 6362 | 7088 | 7854 |    |    |     |   |      |      |      |      |      |   |    |     |
|     | c.   | Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $1/3$ rd rule taking four equal strips.                                                                                                                                                                                                                                                                                              | 7    | L2   | CO2  |    |    |     |   |      |      |      |      |      |   |    |     |

## OR

|     |     |                                                                                                                                                                                                                                                         |   |    |     |   |   |   |     |   |   |    |   |    |     |
|-----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|---|---|---|-----|---|---|----|---|----|-----|
| Q.8 | a.  | Find the real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimals by using Newton-Raphson method.                                                                                                                                         | 6 | L2 | CO2 |   |   |   |     |   |   |    |   |    |     |
|     | b.  | Use Lagrange's interpolation formula to fit a polynomial for the data<br><table border="1"><tr><td>x</td><td>0</td><td>1</td><td>3</td><td>4</td></tr><tr><td>y</td><td>-12</td><td>0</td><td>6</td><td>12</td></tr></table><br>Hence estimate y at x=2 | x | 0  | 1   | 3 | 4 | y | -12 | 0 | 6 | 12 | 7 | L3 | CO2 |
| x   | 0   | 1                                                                                                                                                                                                                                                       | 3 | 4  |     |   |   |   |     |   |   |    |   |    |     |
| y   | -12 | 0                                                                                                                                                                                                                                                       | 6 | 12 |     |   |   |   |     |   |   |    |   |    |     |
|     | c.  | Evaluate $\int_0^{0.3} (1 - 8x^3)^{1/2} dx$ by using Simpson's $3/8$ th rule taking three equal parts.                                                                                                                                                  | 7 | L2 | CO2 |   |   |   |     |   |   |    |   |    |     |

## Module - 5

|     |    |                                                                                                                                                                                      |   |    |     |
|-----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.9 | a. | Use Taylor's series method to find y at $x = 0.1$ considering terms up to third degree given that $\frac{dy}{dx} = x - y^2$ , $y(0) = 1$                                             | 6 | L2 | CO2 |
|     | b. | Using Runge-Kutta method of fourth order, find $y(0.2)$ for the equation $\frac{dy}{dx} = 3x + \frac{y}{2}$ , $y(0) = 1$ taking $h = 0.2$                                            | 7 | L2 | CO2 |
|     | c. | Given $\frac{dy}{dx} = x^2 + y^2$ and the data $y(0) = 1$ , $y(0.1) = 1.1113$ , $y(0.2) = 1.2507$ , $y(0.3) = 1.426$ , find $y(0.4)$ using Milne's predictor and corrector formulae. | 7 | L2 | CO2 |

## OR

|      |    |                                                                                                                                                                                    |   |    |     |
|------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.10 | a. | Using modified Euler's method. Solve $\frac{dy}{dx} = 1 + \frac{y}{x}$ at $x=1.2$ . given $y(1)=2$ taking $h=0.2$                                                                  | 6 | L2 | CO2 |
|      | b. | Using Runge-Kutta method of fourth order, find $y(1.1)$ for the equation $\frac{dy}{dx} = xy^{1/3}$ , $y(1) = 1$                                                                   | 7 | L2 | CO2 |
|      | c. | Given $\frac{dy}{dx} = 2e^x - y$ and the data $y(0)=2$ , $y(0.1)=2.010$ , $y(0.2)=2.040$ and $y(0.3) = 2.090$ , find $y(0.4)$ by using Adams-Bashforth predictor corrector method. | 7 | L2 | CO2 |

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